FRONT-END DEVELEPMENT HACKATHON

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TITLE: Simple Student Activity Tracker Web Application
HACKATHON LEVEL :1

Use Case: Building a Simple Student Activity Tracker Web Application

1. Use Case and Assumptions

This application is designed for students enrolled in training programs or academic courses who need a simple and structured way to track their daily learning activities, assignments, and progress.

• The application is front-end only (HTML, CSS, JavaScript).

• No backend server or database is used.

• Student activity data can be stored in static JSON or browser local storage.

• Users will access the application via mobile or desktop browsers.

2. Problem Statement

Many beginners struggle with complex productivity tools that require backend setup or advanced configurations. There is a need for a simple, beginner-friendly student activity tracker that is lightweight, easy to use, and helps students monitor their learning progress effectively without technical complexity.

3. Objective

To design and develop a simple , responsive, and interactive fron-tend web application that enables students to view activities, mark them as completed, and track overall progress through a clean and user-friendly interface.

4. Key Features

• Home Page displaying application title and welcome message.

• Activity List View showing Activity Name, Short Description, and Current Status.

• Option to mark activities as Completed.

• Automatic Progress Summary (e.g., '3 out of 6 activities completed').

• Simple navigation between Home and Activity sections.

5. UI and Usability Considerations

• Clean and minimal interface design.

• Readable typography with consistent spacing.

• Responsive layout for mobile and desktop devices.

• Clear buttons and intuitive user interaction.

• Color-coded status indicators (Pending / Completed).

6. Technology Used

• HTML – Structure of the web pages.

• CSS – Styling and responsive design.

• JavaScript – Handling interactivity and progress tracking.

• Optional: React (Basic components and state management).

7. Data Handling Method

Activity data will be managed using either a predefined JSON array or stored dynamically using browser Local Storage. When a user marks an activity as completed, the status is updated and progress summary recalculates automatically.

8. High-Level UI Flow / Wireframe Description

Home Page → Click 'View Activities' → Activity List Page → Mark Activity as Completed →

Progress Summary Updates Automatically.

9. Implementation Approach

Step 1: Create basic HTML structure for Home and Activity sections.

Step 2: Apply CSS styling for clean layout and responsiveness.

Step 3: Use JavaScript to load activity data and render dynamically.

Step 4: Add event listeners to update completion status.

Step 5: Calculate and display progress summary in real-time.

10. Output

Student Activity Tracker

Track your learning activities easily

0 out of 6 activities completed

HTML Basics

Learn HTML tags and structure

Pending

Mark Completed

CSS Styling

Design responsive layouts

Pending

Mark Completed

JavaScript Fundamentals

Learn variables and functions

Pending

Mark Completed

Mini Project

Build a small web project

Pending

Mark Completed

Debugging Practice

Fix sample errors

Pending

Mark Completed

Mock Test

Complete practice test

Pending

Mark Completed

11. Expected Outcome

A lightweight and fully functional front-end web application that allows students to easily monitor and manage their learning activities with improved productivity and clarity.

12. Learning Outcome

Students will understand UI design principles, front-end development workflow, basic JavaScript logic, state handling, and responsive web design concepts.

13. Conclusion

The proposed Student Activity Tracker provides a simple yet effective solution for tracking academic progress. By focusing only on front-end technologies, it ensures ease of development and usability while fulfilling all Hackathon Level-1 requirements.